

Ongoing / Previously Discussed Projects

1. Engineering Student Portal (Major Project)

You are building an internal portal for engineering students at **Anzai University**.

Target Users

- Around **5,000 students**
- Mainly Computer Science and Information Technology students

Planned Features

- Attendance tracking
- Class materials and notes
- Classroom allocation
- Technical event management
- Cultural event management
- Suggestion/complaint box

Technology Choice

- Backend planned using **Firebase**

What You Requested

- Detailed requirements document
- System architecture document
- Cost estimation
- Scalability planning
- Future expansion for junior batches

2. Offline LLM + RAG System

Recently you've been discussing building an:

- Offline AI assistant
- Local RAG (Retrieval-Augmented Generation) system
- PDF document question-answering system

Hardware Mentioned

- 16 GB RAM
- RTX 3050 GPU
- Windows

Architecture Being Discussed

- Local LLM through **Ollama**
- Vector database for document retrieval
- PDF ingestion pipeline
- Fully offline querying

You specifically asked whether:

We are using RAG architecture together with an offline LLM from Ollama.

The answer was yes, and we were moving toward a complete implementation guide.

3. ANPR / Computer Vision Project

You were working on a vehicle-related computer vision system using:

- **YOLO**
- Video processing with OpenCV
- Frame extraction
- License plate / vehicle detection workflow

You shared Python code and requested:

- Error fixes
 - Complete code generation
 - Improved frame extraction logic
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4. Drone Building Project

You were designing and assembling a drone.

Topics discussed included:

- Power wiring
- ESC connections

- Battery wiring
- Wire gauge selection
- 14 AWG silicone wire
- Flexible copper wire alternatives
- Using 1.5 sq mm flexible copper wire

You were trying to source suitable wiring locally and verify whether certain wire types could safely connect the battery and ESCs.

5. Programming Practice

You have also worked on:

- C programming
 - Algorithm problems
 - Debugging functions such as Roman numeral conversion
 - General coding error correction
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Your Main Areas of Interest (from our chats)

1. Artificial Intelligence & Local LLMs
2. RAG Systems
3. Computer Vision (YOLO/OpenCV)
4. Full-Stack Development
5. University Management Systems
6. Embedded/Hardware Projects (Drone Building)
7. Software Architecture & Scalability

What I Don't Know

I don't know things such as:

- Your age
- Exact address
- Phone number

- Passwords
- Personal documents
- Academic records
- Financial information

I only know information you've discussed in our conversations and that has been stored as memory for future assistance.

If you want, I can also create a "**Vinod Project Dashboard**" showing:

- All projects
- Current status
- Next tasks
- Technologies used
- Estimated completion roadmap for each project.